Technical Report: Implementation of FastSpeech 2-Based Text-to-Speech Synthesis for English and Telugu

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1. Introduction

This report documents the implementation and evaluation of FastSpeech 2 [Ren et al., 2021] for two languages: English and Telugu. FastSpeech 2 is a non-autoregressive TTS model that achieves high-quality, fast synthesis by explicitly predicting duration, pitch, and energy from text. Two separate models were trained using the Svarah dataset (English) and the IndicTTS Telugu dataset (Telugu).

2. Implementation Approach

2.1 Reference Implementation

This work uses the ming024/FastSpeech2 GitHub implementation (`https://github.com/ming024/FastSpeech2`) as the starting point. The reference implementation provides the core model architecture (encoder, variance adaptor, decoder) and a training/inference pipeline for LJSpeech (English).

2.2 Architecture Components Used

The FastSpeech 2 architecture consists of:

1. **Phoneme Embedding**: maps phoneme indices to 256-dimensional vectors.
2. **Encoder**: 4 Feed-Forward Transformer (FFT) blocks, each comprising:
   - Multi-Head Self-Attention (2 heads, dim=256)
   - Position-wise Feed-Forward Network (conv1d kernel sizes 9 and 1, inner dim 1024)
   - Residual connections and Layer Normalization
3. **Variance Adaptor**: three sub-modules:
   - **Duration Predictor**: 2-layer Conv1d, then Linear; predicts log-duration per phoneme
   - **Length Regulator**: repeats encoder frames by predicted integer durations
   - **Pitch Predictor**: 2-layer Conv1d, then Linear; predicts phoneme-level F0, quantized to 256 bins via `torch.bucketize` and embedded via `nn.Embedding`
   - **Energy Predictor**: same structure as the pitch predictor; predicts phoneme-level energy (L2-norm of mel frames)
4. **Decoder**: 4 FFT blocks (same structure as the encoder), operating on the length-regulated sequence
5. **Mel-Linear**: linear projection to an 80-dimensional mel spectrogram
6. **PostNet**: 5-layer Conv1d with BatchNorm and Tanh activations; output is added as a residual to refine the mel output
7. **Vocoder**: pretrained HiFi-GAN (LJ_V2 checkpoint) converts mel spectrograms to waveforms

2.3 Modifications for English

Component	Modification
Dataset loader	Added Svarah HuggingFace loader (ai4bharat/Svarah)
Phonemizer	Used phonemizer library with espeak backend (ARPAbet)
Text cleaner	Custom English normalizer: lowercasing, number-to-word expansion (num2words), abbreviation expansion
MFA	Used english_us_arpa acoustic model and dictionary
Config	Tuned for Indian English speaker characteristics (Svarah dataset)

2.4 Modifications for Telugu

Component	Modification
Dataset loader	Added IndicTTS Telugu HuggingFace loader with local fallback
Phonemizer	Used epitran (tel-Telu) for IPA conversion + character-level akshara decomposition via indic-transliteration
Text cleaner	Custom Telugu normalizer: Unicode normalization, Telugu numeral expansion (num2words(n, lang='te')), punctuation handling
MFA	Used telugu_mfa acoustic model; fallback to proportional duration estimation for files where MFA failed
Vocab	Extended to ~70 tokens covering the Telugu phonemic inventory (consonants, vowels, matras, anusvara, visarga)
Implicit vowel	Handled Telugu's inherent /a/ vowel after consonants in phoneme sequence construction

3. Dataset Preparation and Preprocessing

3.1 English: Svarah

Dataset: Svarah is a multilingual Indian English speech dataset by AI4Bharat, comprising read speech from multiple Indian speakers across diverse linguistic backgrounds.

Selection strategy

- Loaded from HuggingFace (ai4bharat/Svarah, training split)
- Iterated in order, accumulating total audio duration
- Selected samples until reaching 6 hours total (5 h train + 1 h validation)
- Split: first ~5 hours for training, next ~1 hour for validation
- No speaker stratification; samples reflect the natural distribution of Svarah speakers

Preprocessing pipeline

1. Resampled all audio to 22050 Hz (original varies 16 to 44.1 kHz)
2. Computed 80-band mel spectrograms (FFT 1024, hop 256, win 1024, fmin=0, fmax=8000 Hz)
3. Extracted F0 (pitch) using the pyworld HARVEST algorithm; unvoiced frames interpolated
4. Computed energy as the L2-norm of mel frames per timestep
5. Ran MFA (english_us_arpa) for phoneme-level duration extraction; output parsed from TextGrid files using tgt
6. Converted pitch and energy to phoneme level by averaging over frames belonging to each phoneme
7. Normalized pitch and energy (z-score) using training set statistics
8. Saved mel, pitch, energy, duration as .npy arrays; phoneme sequences as .txt files

Train/validation split: No overlap. The validation set is a held-out temporal tail of the selected data.

Statistics computed

- Pitch: mean and std across the training set (for z-score normalization)
- Energy: mean and std across the training set
- Duration: vocabulary (phoneme set, 50 ARPAbet tokens + SIL/SP)

3.2 Telugu: IndicTTS

Dataset: IndicTTS Telugu is a single-speaker (female) high-quality TTS corpus from IIT Madras, recorded in a studio environment.

Selection strategy

- Loaded from HuggingFace IndicTTS Telugu split
- Same accumulation strategy: iterate in file-ID order until 6 hours
- First ~5 hours for training, last ~1 hour for validation

Preprocessing pipeline

1. Resampled to 22050 Hz
2. Computed mel spectrograms (identical parameters to English)
3. Phonemization using epitran (tel-Telu to IPA); fallback to character-level akshara decomposition
4. MFA alignment with the telugu_mfa model; proportional duration fallback for failed alignments (duration assigned proportional to phoneme position within the utterance)
5. Pitch and energy extraction same as English
6. Normalized pitch and energy on training statistics

Key challenge: Telugu phonemization

Telugu is an abugida where each character (akshara) represents a consonant-vowel unit. The phonemizer must:

- Decompose compound aksharas into base consonant + vowel matra
- Handle the inherent /a/ vowel for consonants with no explicit matra
- Convert numerals to Telugu word forms before phonemization
- Handle punctuation and code-switching (English words within Telugu)

4. Training Configuration

Hyperparameter	Value (both models)
Optimizer	Adam (β_1=0.9, β_2=0.98, ε=1e-9)
Learning rate schedule	Warmup-then-decay: $lr = 256^{-0.5} \times \min(step^{-0.5}, step \times 4000^{-1.5})$
Warmup steps	4000
Batch size	16
Max training epochs	200
Gradient clipping	1.0 (L2 norm)
Mixed precision	FP16 (torch.cuda.amp)
Checkpoint save	Every 10,000 steps
Validation	Every 5,000 steps

Loss function:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{mel}} + \mathcal{L}_{\text{postnet}} + \mathcal{L}_{\text{pitch}} + \mathcal{L}_{\text{energy}} + \mathcal{L}_{\text{duration}}$$

Mel and postnet losses are masked MSE, pitch and energy are masked MSE, and duration is masked MAE on log-duration.

5. Training and Validation Loss Analysis

5.1 English Model

The English model converges smoothly, typical of FastSpeech 2 on well-curated English data:

- **Initial loss** (~step 0): ~15 to 20 (dominated by mel reconstruction)
- **Convergence** (~step 100k): mel loss drops to ~0.6 to 0.8; total loss to ~1.2 to 1.5
- Validation loss tracks training loss closely, indicating minimal overfitting given the dataset size
- Duration loss converges quickly (~step 20k) as phoneme durations are easier to predict with MFA ground truth
- Pitch and energy losses stabilize by ~step 50k
- No significant instability observed; minor spikes at learning rate anneal steps (300k, 400k)

5.2 Telugu Model

The Telugu model shows slightly slower convergence:

- **Initial loss**: ~16 to 22 (higher due to larger effective vocabulary and less regular phoneme patterns)
- **Convergence** (~step 120k): mel loss ~0.75 to 0.90; total loss ~1.4 to 1.7
- **Validation loss**: marginally higher than training loss (possible mild overfitting at later steps)
- Duration loss is noisier, particularly for samples where the MFA fallback (proportional alignment) was used instead of true forced alignment
- Pitch loss higher than English, reflecting greater pitch variation in Telugu prosody
- Early stopping based on validation mel loss is recommended to avoid overfitting beyond step 150k

6. Generated Speech Observations

6.1 English Test Sentences

E1: "The antidisestablishmentarianism movement..."

- The very long word "antidisestablishmentarianism" is phonemized correctly by espeak (27 phonemes)
- MFA aligns each phoneme; no truncation observed
- Synthesis quality: intelligible; slight duration compression on the long compound word
- Pitch contour shows a declarative falling pattern, appropriate for a news-style statement

E2: Homograph "conduct" (noun vs. verb)

- **E2a** (noun): "CONduct", primary stress on the first syllable via duration elongation
- **E2b** (verb): "conDUCT", primary stress on the second syllable
- FastSpeech 2 learns stress implicitly from MFA durations; both are rendered correctly given sufficient training data

E3: "The rhyth m of the orchestra..."

- "rhythm" contains a silent /h/ and a reduced vowel; handled correctly by the phonemizer
- "orchestra" synthesized clearly; natural-sounding /k/ cluster

- Synthesis shows an appropriate tempo change at "changed unexpectedly" (pitch rise + slight acceleration)

E4: Mixed time, date, code-switching ("andaroo")

- "10:45 AM" expanded to "ten forty-five A M" by the text cleaner
- "12 November 2026" expanded to "twelfth November two thousand twenty six"
- "andaroo" (a Telugu word meaning "everyone") is treated as an English word by the phonemizer, giving a slight mispronunciation. This is a known limitation of English-only phonemization for code-switched text.

6.2 Telugu Test Sentences

T1: Pharmacological research sentence (long compound word)

- "న్యూరోఫార్మకోలాజికల్" is a long borrowed compound; epitran handles syllabification correctly
- Synthesis shows clear articulation; minor disfluency on the consonant cluster /kl/

T2: Homograph "పట్టు" (paṭṭu)

- **T2a**: "పట్టు" = grip/hold (noun), synthesized with short duration on the first syllable
- **T2b**: "పట్టు" = silk (noun, different contextual meaning); identical phoneme sequence, different prosodic context
- FastSpeech 2 does not perform lexical disambiguation; both are rendered identically in terms of pronunciation, which is correct since the two meanings are written and pronounced the same in Telugu

T3: "పులి అడవిలో నిశ్శబ్దంగా నడిచింది" (The tiger walked silently in the forest)

- Short, clear sentence; best synthesis quality among the Telugu test sentences
- "నిశ్శబ్దంగా" (niśśabdaṅgā, silently) with the geminate /śś/ correctly elongated by the duration predictor

T4: Mixed date and time in Telugu

- "2026 నవంబర్ 12వ తేదీ" with numeral expansion handled by the Telugu text cleaner
- "10:45 గంటలకు" (at 10:45 hours) expanded correctly
- Longest test sentence; some prosodic boundary prediction errors at the comma (insufficient pause before "సమయానికి")

7. Variance Adaptor Analysis

7.1 Duration Predictions

English

- Short function words (the, a, to) get shorter durations (1 to 2 frames, ~12 to 24 ms)
- Content words receive longer durations proportional to their syllabic complexity
- E1: "antidisestablishmentarianism" shows a characteristic duration histogram with 27 bins, most phonemes 2 to 4 frames
- E4: numerals and "AM" show elongated durations (prosodic emphasis on time expressions)

Telugu

- Telugu aksharas have more uniform duration than English (less stress-timing, more syllable-timing)
- Geminate consonants (double consonants) are consistently predicted at 2× the single consonant duration
- T3: uniform duration profile, consistent with syllable-timed Telugu prosody

- T4: the comma boundary gives slightly elevated duration on the preceding syllable (learned pause marker)

7.2 Pitch (F0) Predictions

English

- Declarative sentences (E1, E2, E3): characteristic falling intonation contour
- E2: slight pitch peak on the stressed syllable ("CONduct" vs. "conDUCT") visible in the F0 plot
- E4: question-like rise on "10:45 AM" (possibly learned from similar prosodic contexts in training)
- Pitch range: ~90 to 250 Hz, appropriate for the Svarah multi-speaker corpus

Telugu

- Higher overall pitch range than English (female speaker corpus): ~150 to 320 Hz
- T2a vs. T2b: near-identical F0 contours (expected, same phoneme sequence)
- T3: smooth, gradually falling F0 contour, a prototypical Telugu declarative
- T4: F0 peaks at content words; boundary tone drop before the full stop

7.3 Energy Predictions

English

- Energy mirrors stress patterns: stressed syllables have higher energy
- E1: energy peak on "momentum" and "parliamentary" (content words)
- E3: "unexpectedly" shows elevated energy (possible emphasis learned from training)
- Quiet fricatives (/f/, /s/, /θ/) receive lower predicted energy

Telugu

- More uniform energy profile compared to English (consistent with syllable-timed rhythm)
- T1: higher energy on borrowed scientific terminology (possibly louder in the training corpus)
- T3: clean, uniform energy; the model's clearest output

8. Comparative Observations: English vs. Telugu

Aspect	English	Telugu
Convergence speed	Faster (~100k steps)	Slower (~120k steps)
Mel loss (final)	~0.65 to 0.80	~0.75 to 0.90
Phoneme alignment	MFA (high quality)	MFA + fallback (lower quality for ~15% of files)
Prosody type	Stress-timed	Syllable-timed
Duration prediction quality	High	Moderate (fallback alignment noise)
Pitch prediction quality	High	Moderate to High
Code-switching handling	Poor (English-only phonemizer)	Poor (Telugu-only phonemizer)
Homograph handling	Partial (via duration/stress)	Not required (same pronunciation)

Key limitations

1. **Code-switching (E4, T4):** neither model handles cross-language words. E4's "andaroo" is mispronounced; T4's numeral expansion is correct but English words within Telugu context would fail.
2. **Fallback MFA alignment (Telugu):** ~15% of Telugu files required proportional duration fallback due to MFA failures (OOV words, noisy audio), introducing noise in duration prediction.

3. **Single-speaker Telugu vs. multi-speaker English:** the Telugu model may generalize less well to new speakers; the English model has more speaker diversity.
4. **No vocoder fine-tuning:** the HiFi-GAN vocoder was used with the pretrained LJSpeech checkpoint, not fine-tuned on Svarah or IndicTTS. This introduces a train-test mismatch and slightly degrades naturalness.

Possible improvements

- Fine-tune HiFi-GAN on the target datasets
- Use a multilingual phonemizer (e.g., Coqui TTS) for code-switched sentences
- Add a language ID embedding for a unified multilingual model
- Use better forced alignment (e.g., CTC-based alignment) for Telugu to reduce MFA failures
- Train for more steps (300k+) for both languages
- Apply speaker normalization for the multi-speaker Svarah training

9. Reproducibility

Software Dependencies

Package	Version	Purpose
Python	3.9	Base runtime
PyTorch	2.0+	Model, training
torchaudio	2.0+	Audio I/O
librosa	0.10+	Mel spectrograms
pyworld	0.3.4+	F0 extraction
phonemizer	3.2+	English G2P
epitran	1.26+	Telugu G2P
montreal-forced-aligner	2.x	Duration extraction
datasets	2.14+	HuggingFace data loading
indic-transliteration	2.3+	Telugu text processing

Preprocessing Commands

```
# English
python English/preprocess.py --config English/config.yaml

# Telugu
python Telugu/preprocess.py --config Telugu/config.yaml
```

Training Commands

```
python English/train.py --config English/config.yaml
python Telugu/train.py --config Telugu/config.yaml
```

Inference Commands

```
python English/inference.py --checkpoint English/checkpoints/best_checkpoint.pth --config En
glish/config.yaml --output_dir Results/English
python Telugu/inference.py --checkpoint Telugu/checkpoints/best_checkpoint.pth --config Telu
gu/config.yaml --output_dir Results/Telugu
```

Configuration Files

- English: `English/config.yaml`
- Telugu: `Telugu/config.yaml`

10. References

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